Virtualization Basics

Level 0–1 IT Basics - 04 Virtualization

Security Level:

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Origin of Virtualization

Virtualization development history

- In 1960s, the virtualization technology was applied on mainframe computers.
- In 1999, logical partition was used on mid-range computers.
- In **2000**, virtualization on the x86 platform emerged.
- In **2001**, x86 virtualization was used on servers.
- In **2019**, Arm virtualization was used on Huawei TaiShan servers**.**

Virtualization driving forces

- CPUs are getting faster, exposing higher requirements on hardware.
- Intel, AMD, and Arm add virtual instructions to the CPU.
- Enterprises need to reduce costs.
- Environment protection becomes more and more important.
- Service pressure constantly increases.

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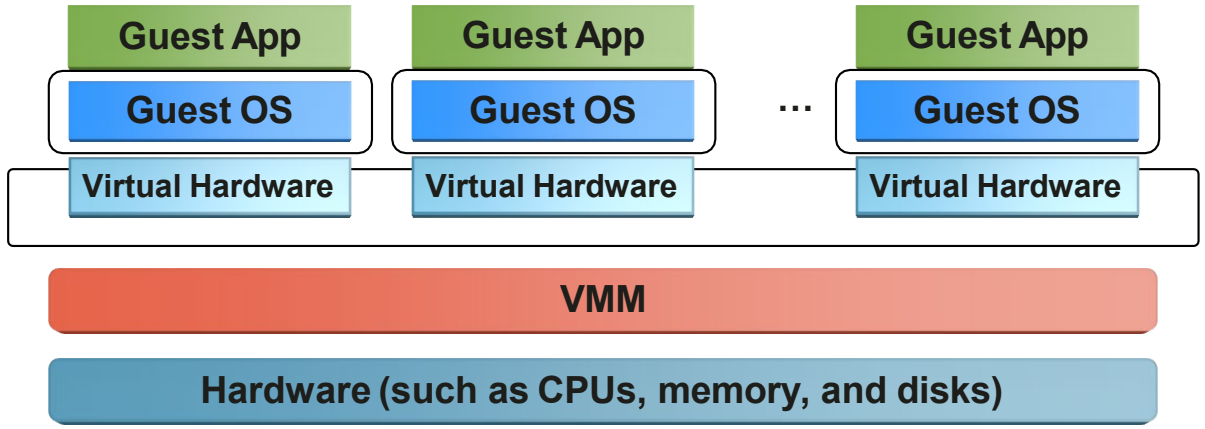
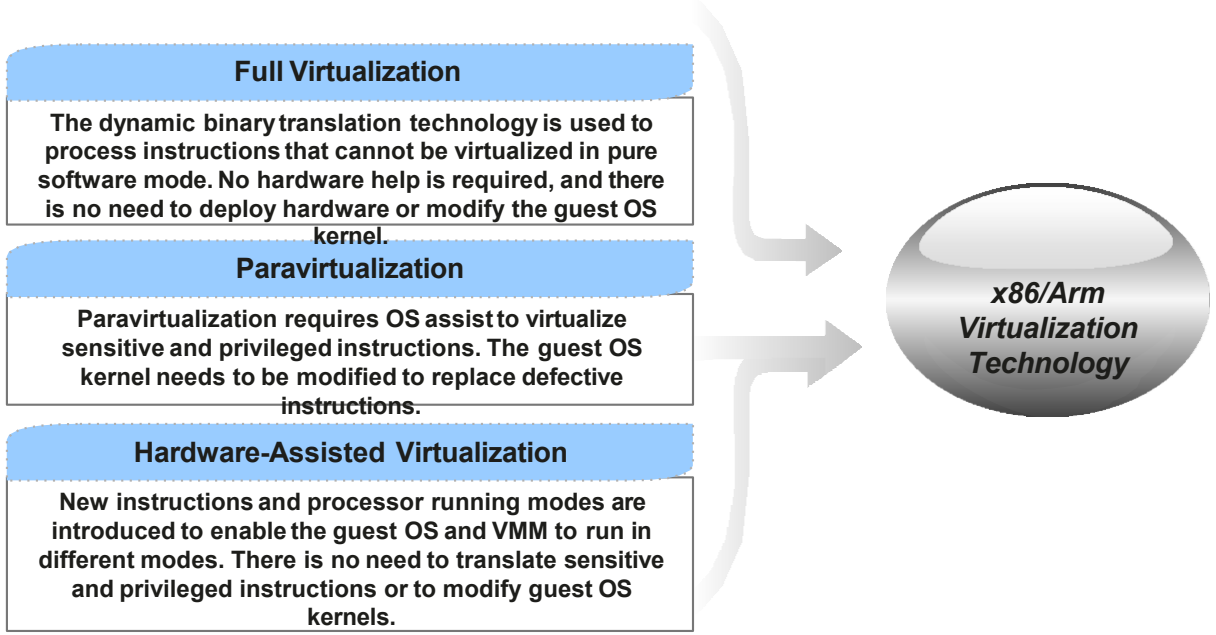
What Is Virtualization?

Starting from server virtualization

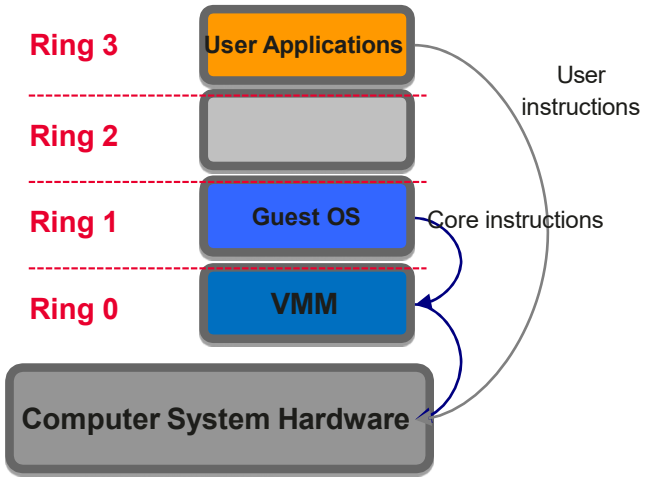
- Server virtualization refers to simulating multiple independent virtual hardware systems that have the same instruction sets as hardware on an x86 hardware platform. Virtual hardware systems can run different OSs (guest OSs). These guest OSs access actual physical resources through virtual machine monitor (VMM).

- VMM is a software layer between the computer hardware and guest OS. VMM has the highest privilege level, and manages and isolates multiple VMs running at the upper layer to provide a secure and independent runtime environment for these VMs. The virtualized hardware environment of each VM is irrelevant to the real hardware, such as monitors, hard disks, CPUs, memory, and NICs.

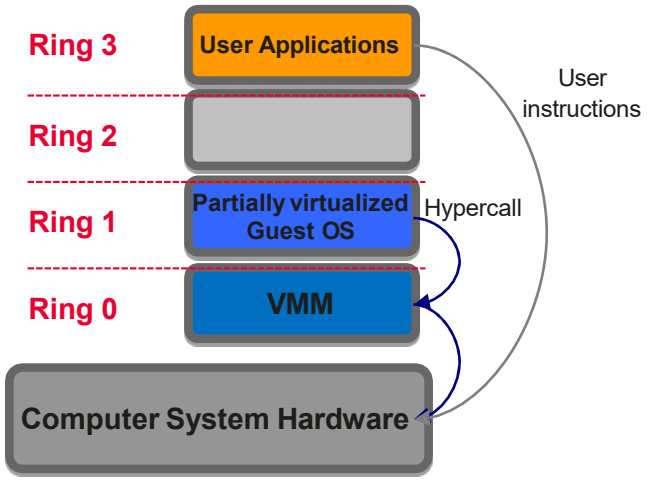




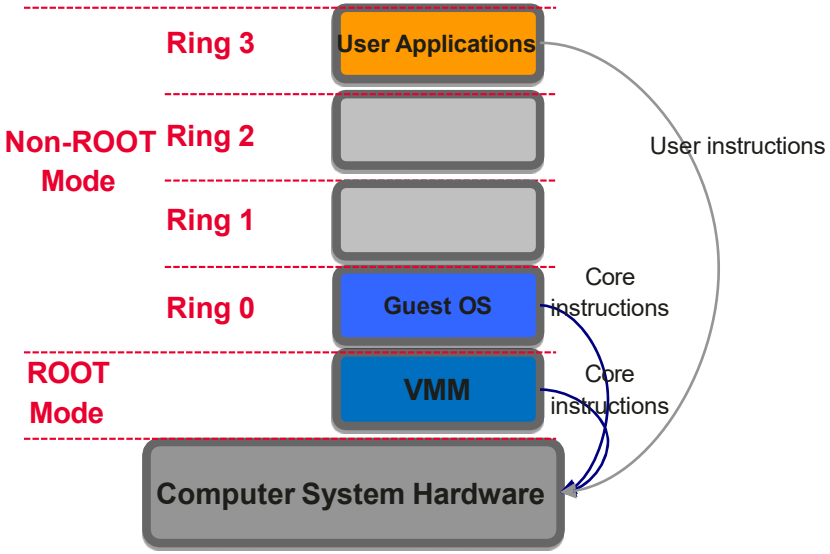
Origin of Virtualization – Classification of Virtualization Technologies



Full Virtualization



Paravirtualization



Hardware-Assisted Virtualization

Comparison Among Three Computing Virtualization Technologies

	Full Virtualization	Paravirtualization	Hardware-Assisted Virtualization
Technology	• Binary translation • Hypervisor implements the virtualization of CPU, memory, and device I/O.	• Hypercall • Hypervisor implements the virtualization of CPU and memory, and the administrative OS implements the virtualization of device I/O.	• Intel-VT and AMD-V • The CPU uses the VT-x technology, the memory uses the EPT technology, and the device I/O uses the VT-d or VT-c technology.
Whether to Modify the Guest OS and Compatibility	• No need to modify the guest OS, optimal compatibility	• Need to modify the guest OS, **poor compatibility**	• No need to modify the guest OS, optimal compatibility
Typical Virtualization Technology	• VMware, KVM	• Xen	• VMware, KVM

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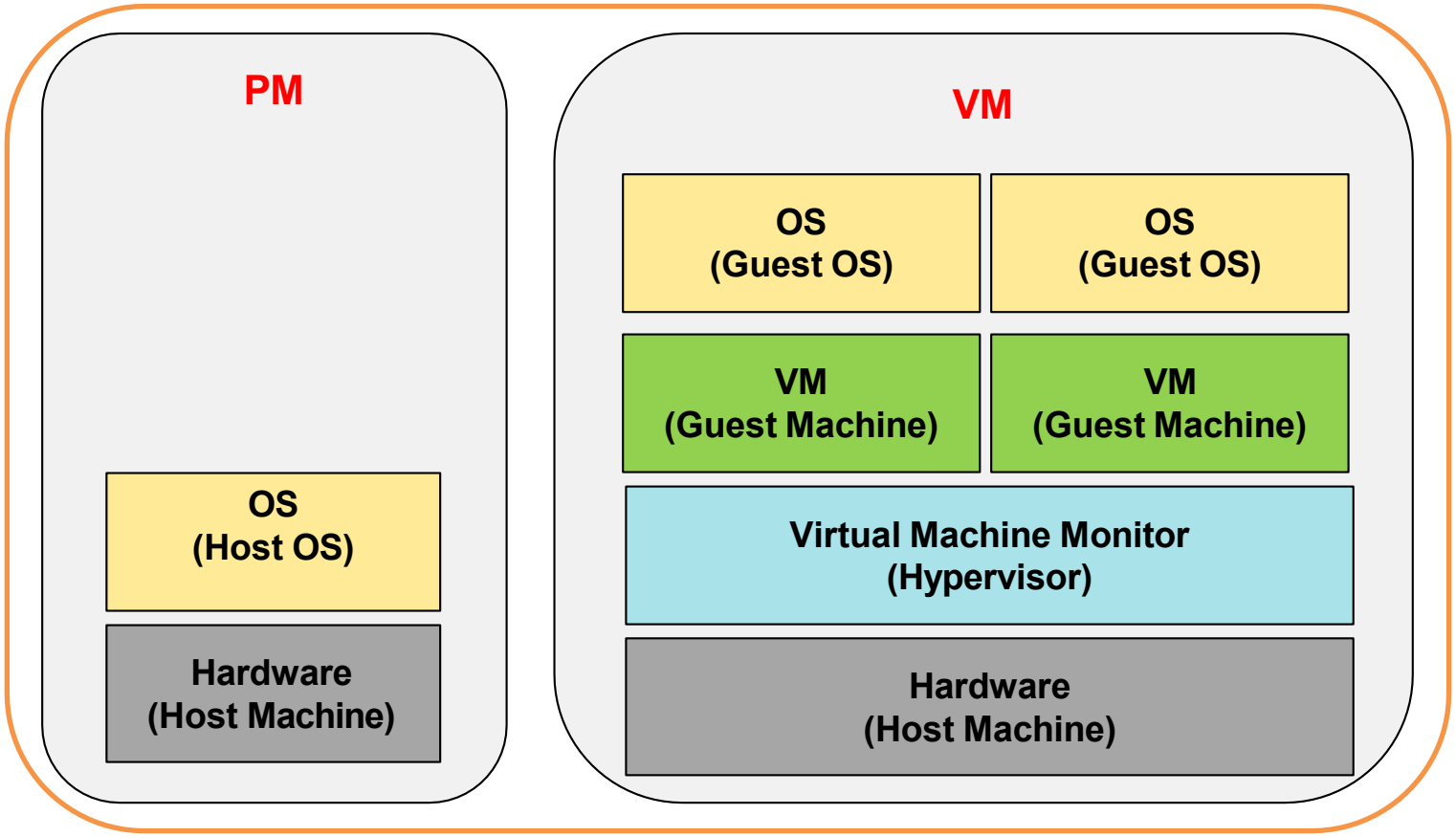
Virtualization Basics – Important Technical Concepts



Guest OS:
an OS running on a VM

Guest Machine:
virtualized VM

Hypervisor:
virtualization software layer or VMM

Host OS:
OS running on physical machines (PMs)

Host Machine:
PM

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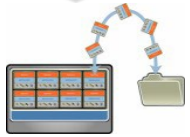
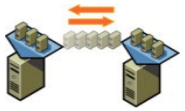
Virtualization Basics – Key Technologies of Virtualization

Partition:
Multiple VMs can run on a single physical server simultaneously.

Isolation:
Multiple VMs are isolated from each other on the same PM.

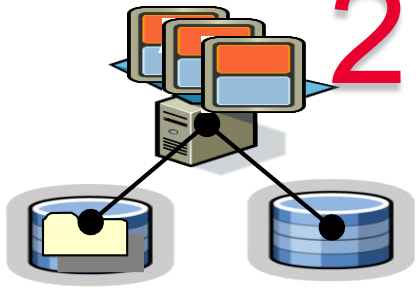
Encapsulation:
The entire VM is stored in files, and can be moved and copied by moving and copying these files.





Hardware Independence: VMs can run on any server without modification.

1 Key Features of Server Virtualization



2 Live Migration of Shared Storage

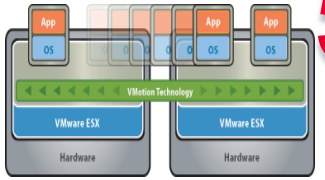
- No VM downtime
- Support for fibre channel

3 High Availability (HA)

- No requirement for exclusive stand-by hardware
- No cost or complexity of cluster software
- Cost-effective and HA for all applications

4 Automatic Allocation of Dynamic Resources

- Apps with higher service priorities preferentially guaranteed
- System management efficiency dynamically improved
- Simplified system engineering based on service planning



5 VM Live Migration

- System capacity expansion with zero downtime
- Load balancing among servers in a distributed system
- Hardware maintenance with zero downtime
- Hitless upgrade

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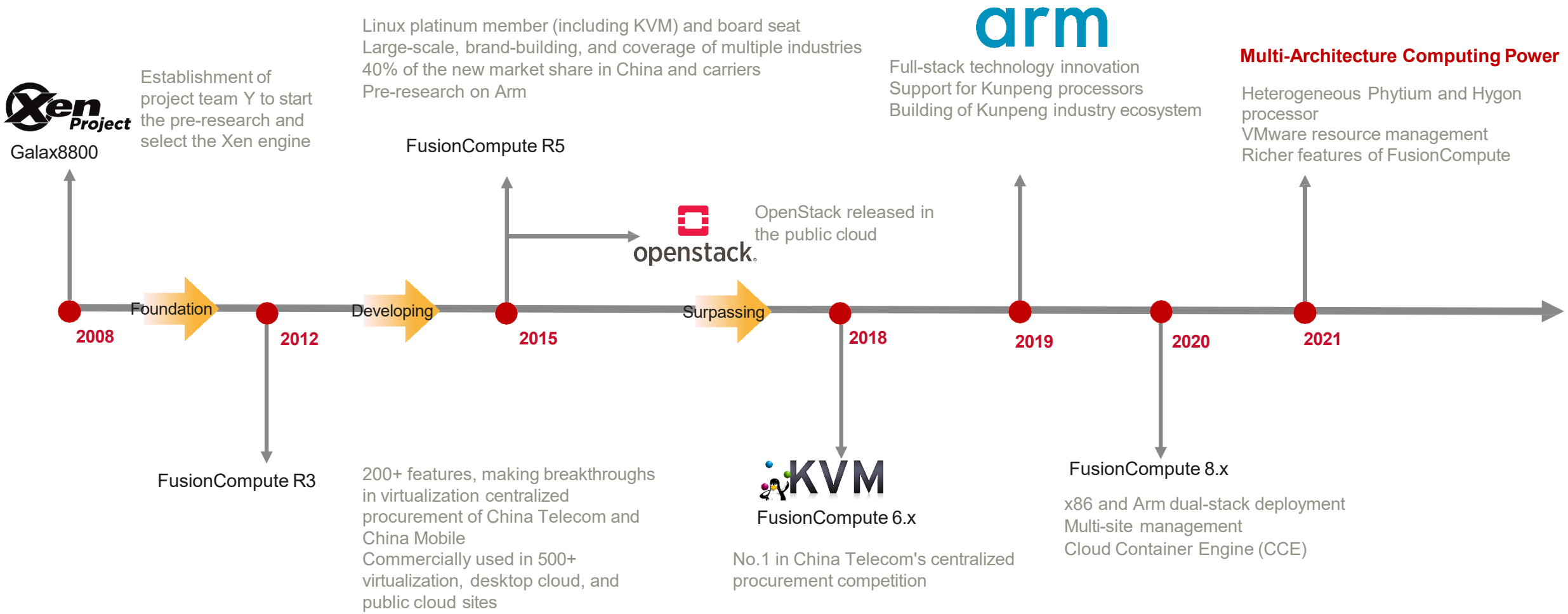
Virtualization Technology – Huawei FusionCompute



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Review of Key Points

Classification of Virtualization Technologies:

- Full virtualization

Guest OSs are separated from underlying hardware, and the hypervisor converts the code invoked by the guest OSs to the underlying hardware. Full virtualization is compatible with guest OSs and does not require the guest OSs to be modified.
Typical cases: VMware Workstation, early-stage ESX servers, and Microsoft Virtual Server

- Paravirtualization

Specific virtualization instructions are added to guest OSs. Then the instructions can directly use the hypervisor to invoke hardware resources, reducing the performance overhead introduced by the hypervisor to convert instructions.
Typical cases: Microsoft Hyper-V, VMware vSphere, and Citrix XenServer

- Hardware-assisted virtualization

A new instruction set and processor running mode are added to the CPU, so that guest OSs can directly invoke hardware resources.
Typical cases: Intel VT and AMD-V

Key Technologies of Virtualization:

① Key features of server virtualization
  a) Partition: Multiple VMs can run on a single PM simultaneously.
  b) Isolation: Multiple VMs are isolated from each other on the same PM.
  c) Encapsulation: The entire VM execution environment is encapsulated in a separate file.
  d) Hardware independence: VMs can run on any PMs without being modified.
② Live migration of shared storage
③ Automatic allocation of dynamic resources
④ VM live migration
⑤ Performance optimization technologies: SR-IOV NIC passthrough, hugepages, NUMA, and DPDK

Basic Concepts

① Guest OS: an OS running on a VM
② Guest machine: virtualized VM
③ Hypervisor: virtualization software layer or VMM
④ Host OS: OS running on PMs
⑤ Host Machine: PM

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